

A Trust-Based Dynamic Routing Framework for Enhancing Security in Software-Defined Networks

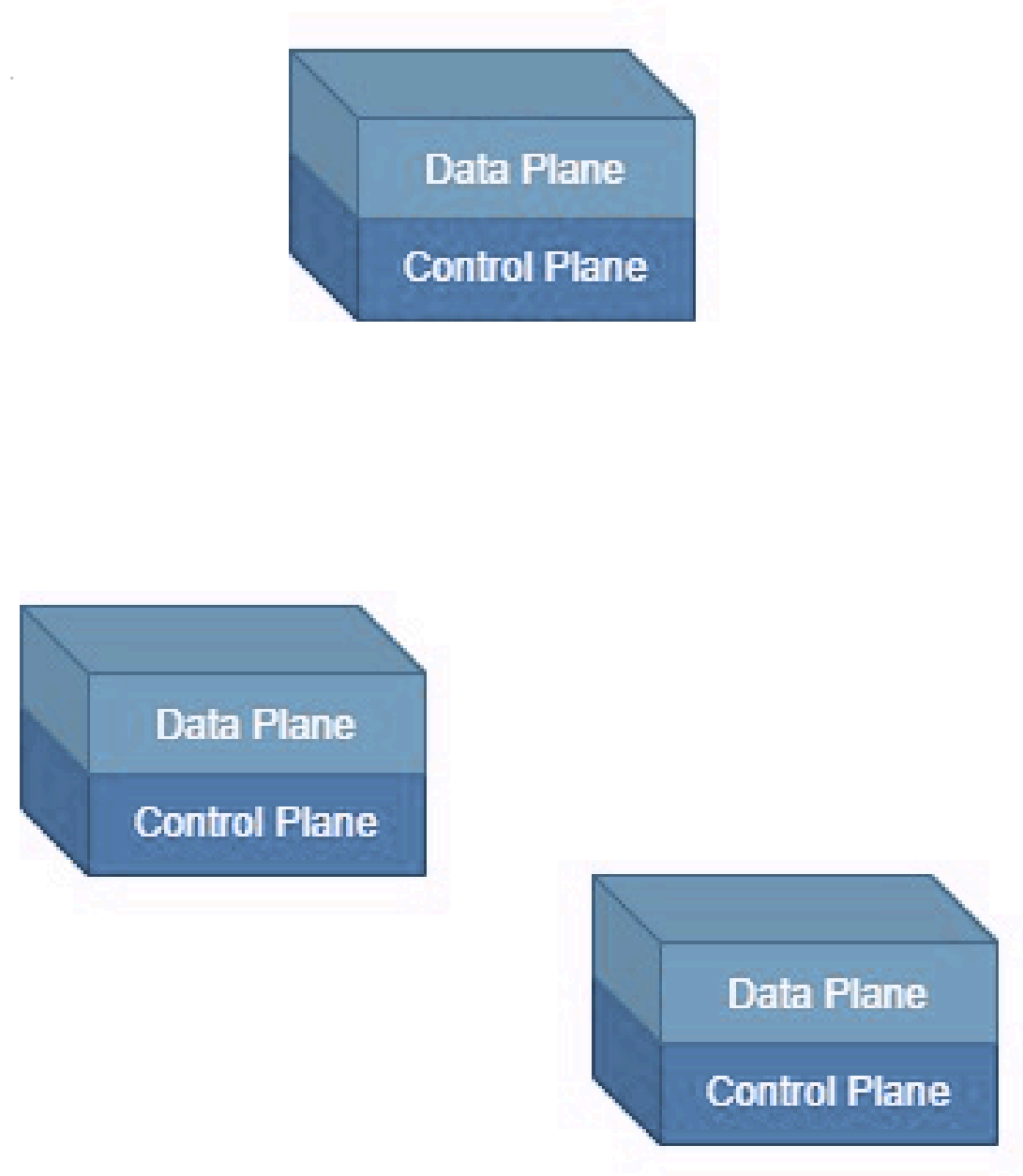
CNS SCE



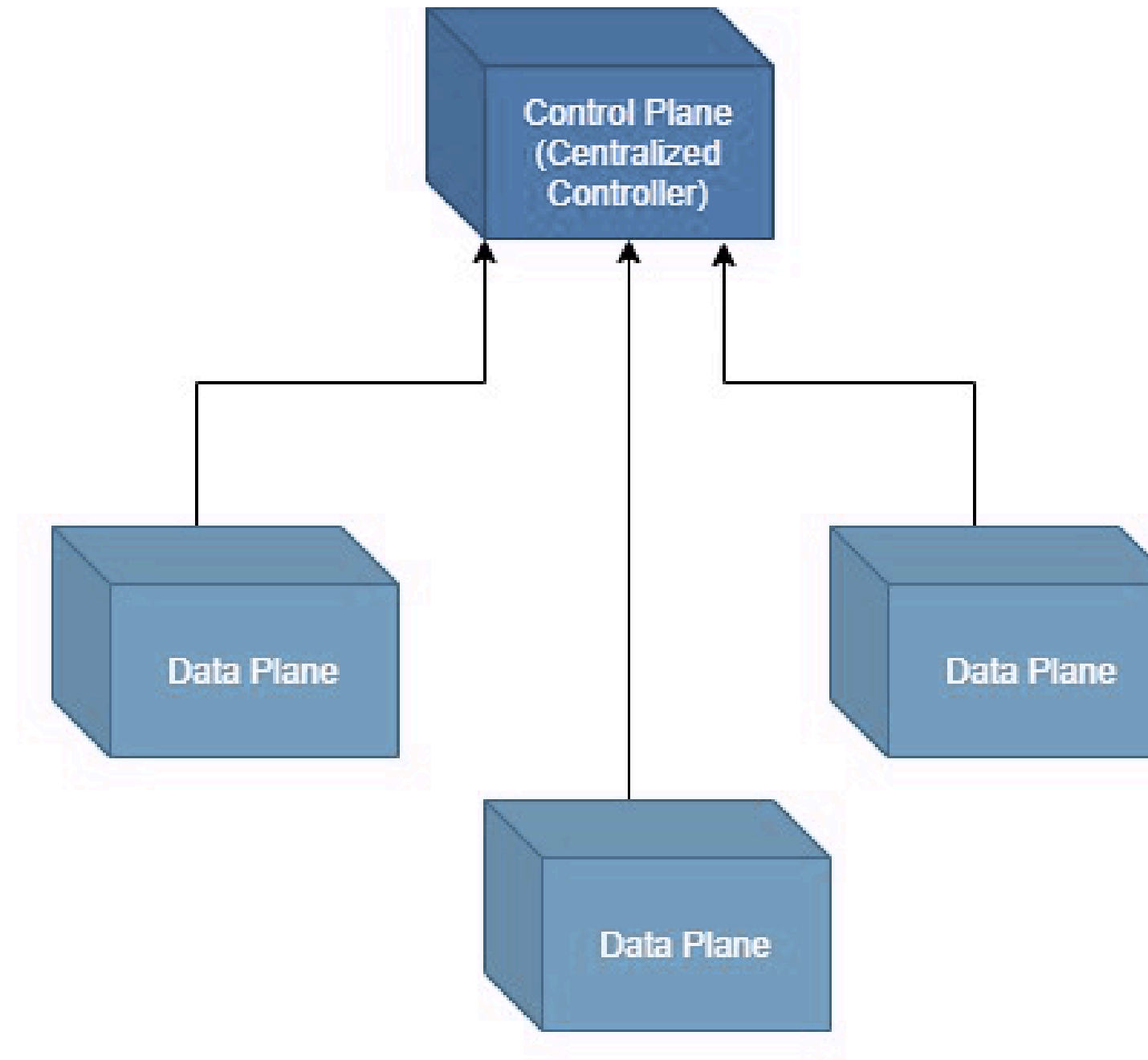
Name	Roll no	PRN
Manas Kulkarni	311058	22311608
Samiksha Nalawade	311063	22311696
Rajlakshmi Desai	311065	22311732

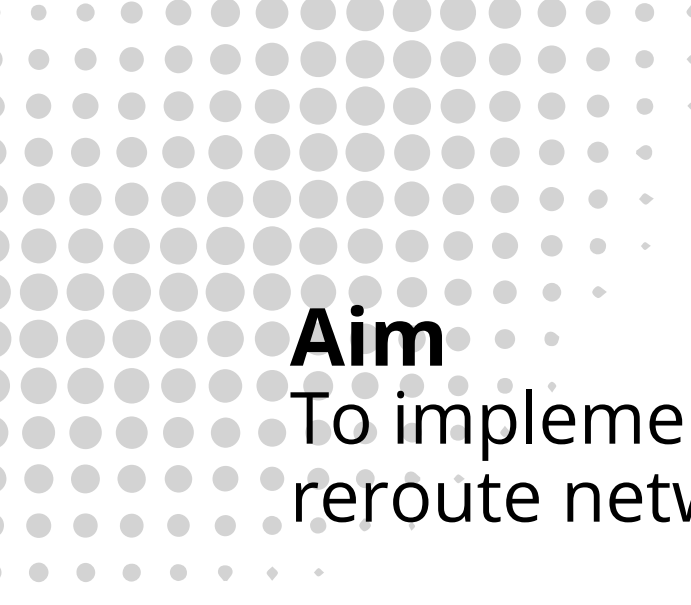
Project Guide: Prof. Dipti Pandit

Traditional Network



Software Defined Network





Aim

To implement a system that can detect malicious switches by trust calculation and automatically reroute network traffic through reliable paths to ensure secure and efficient communication.

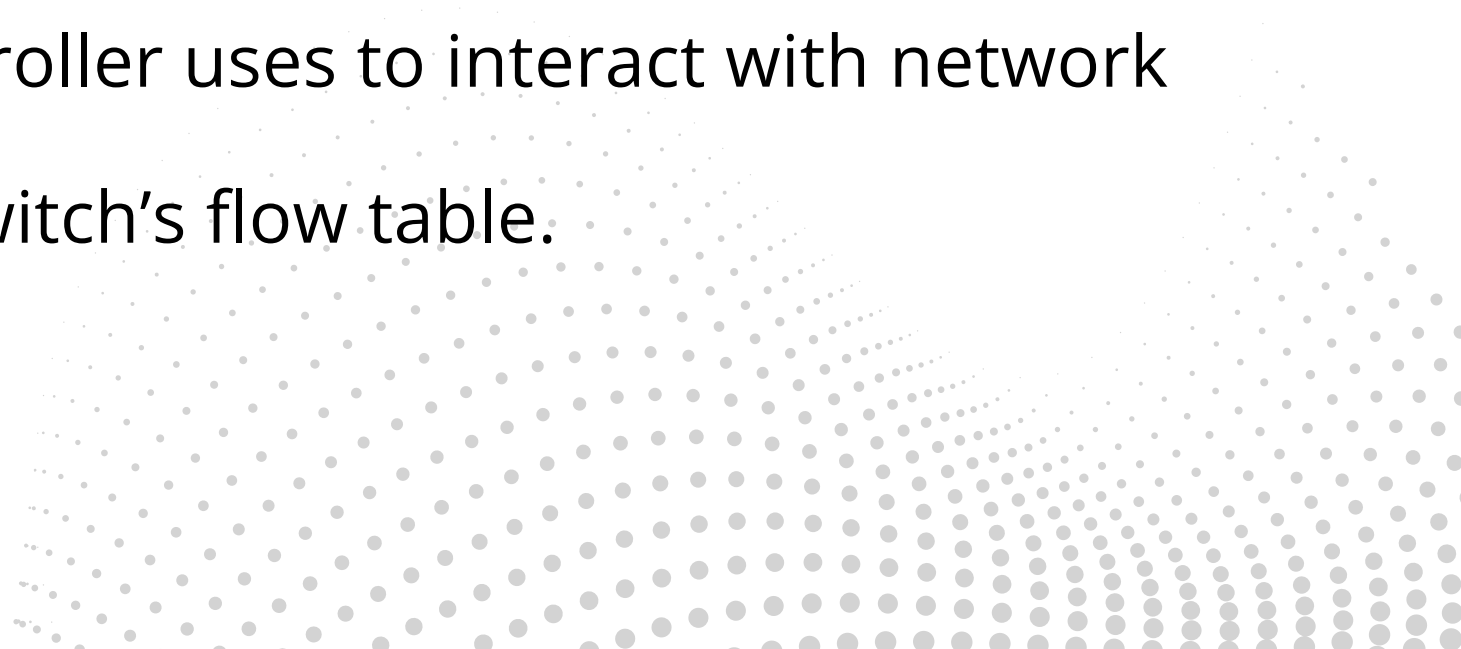
SDN

- Software-Defined Networking (SDN) is a network architecture that separates the control plane from the data plane.
- Better flexibility and scalability
- Simplified network management

RYU Controller

- Ryu is a software-based SDN controller framework written in Python.
- Used to control and manage network devices in SDN.
- Application to perform trust-based routing is extended on this framework

OpenFlow

- OpenFlow is a communication protocol in SDN that controller uses to interact with network switches.
 - It helps controller install and manage flow rules in the switch's flow table.
- 

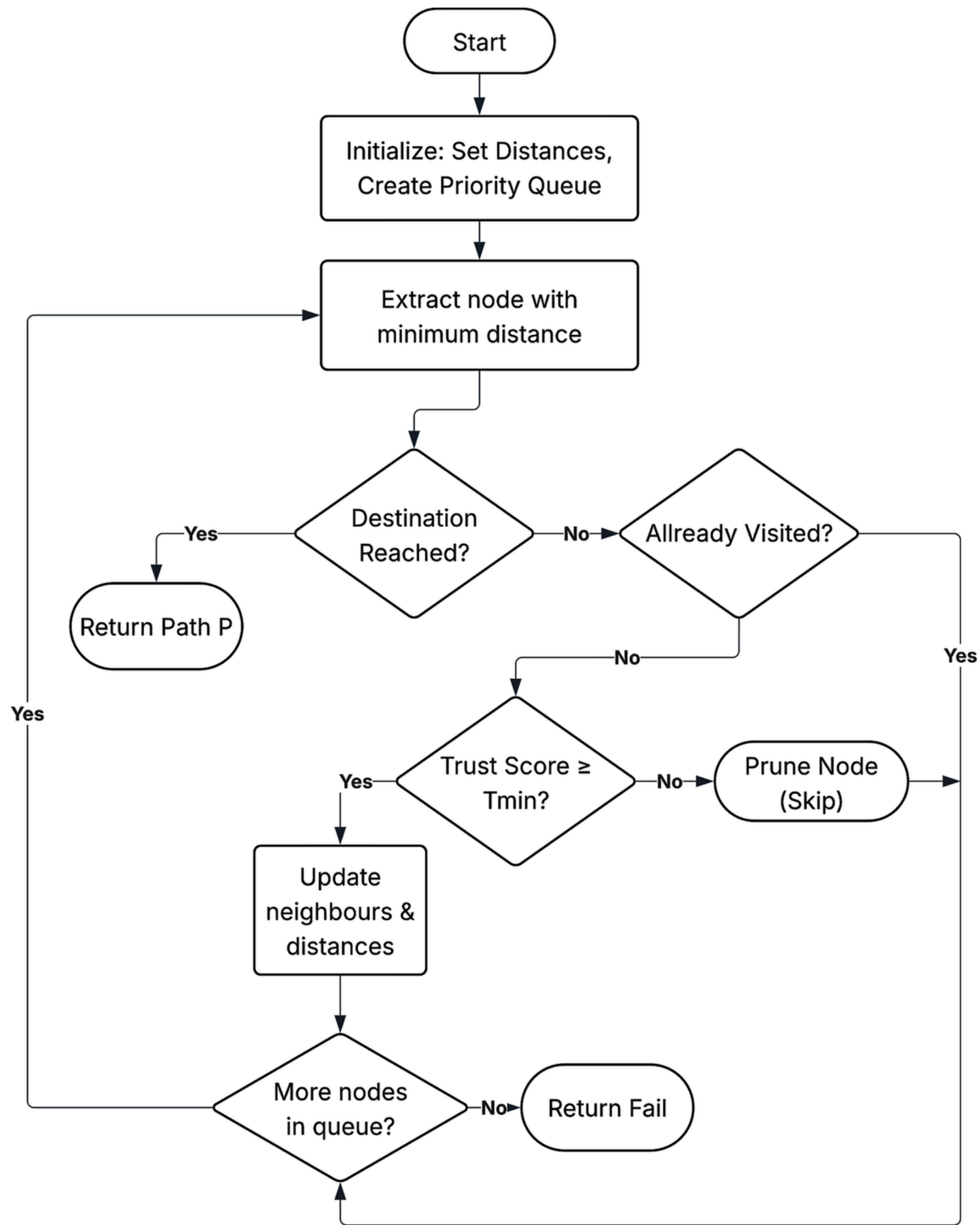
Literature Survey

Paper Name	Year	Summary of Paper	Research Gap
Enhancing east-west interface security in heterogeneous SDN via blockchain	2025	Proposes a blockchain-based security framework for the east-west interface (inter-controller communication) in heterogeneous SDN environments. It utilizes Ethereum smart contracts to handle mutual authentication, encryption, and decentralized access control to mitigate threats like DDoS and false data injection between diverse controllers.	This framework specifically targets the Control Plane's east-west interface (securing controller-to-controller interactions) rather than the Data Plane; it does not propose a mechanism for trust-based dynamic routing of user traffic flows.
Secure Routing in SDN-Enabled 5G Networks: A Trust-Based Model	2025	Presents a dynamic routing model for 5G SDN that calculates trust using behavioral metrics (packet delivery ratio, latency) to mitigate blackhole and greyhole attacks, incorporating a trust aging mechanism.	The model is tailored specifically for 5G infrastructures and relies on performance metrics; your framework could offer a broader application (general SDN) or use deeper trust metrics (e.g., subjective logic) beyond just packet behavior.
Securing the SDN Data Plane in Emerging Technology Domains: A Review	2025	A comprehensive review categorizing SDN data plane security. It identifies "Trust-Based Routing" as an emerging domain that uses node trustworthiness to compute secure end-to-end paths.	As a review paper, it identifies the gap rather than solving it; it explicitly states that trust-based routing is "undergated" and lacks a comprehensive, extensible framework, validating the need for specific research.

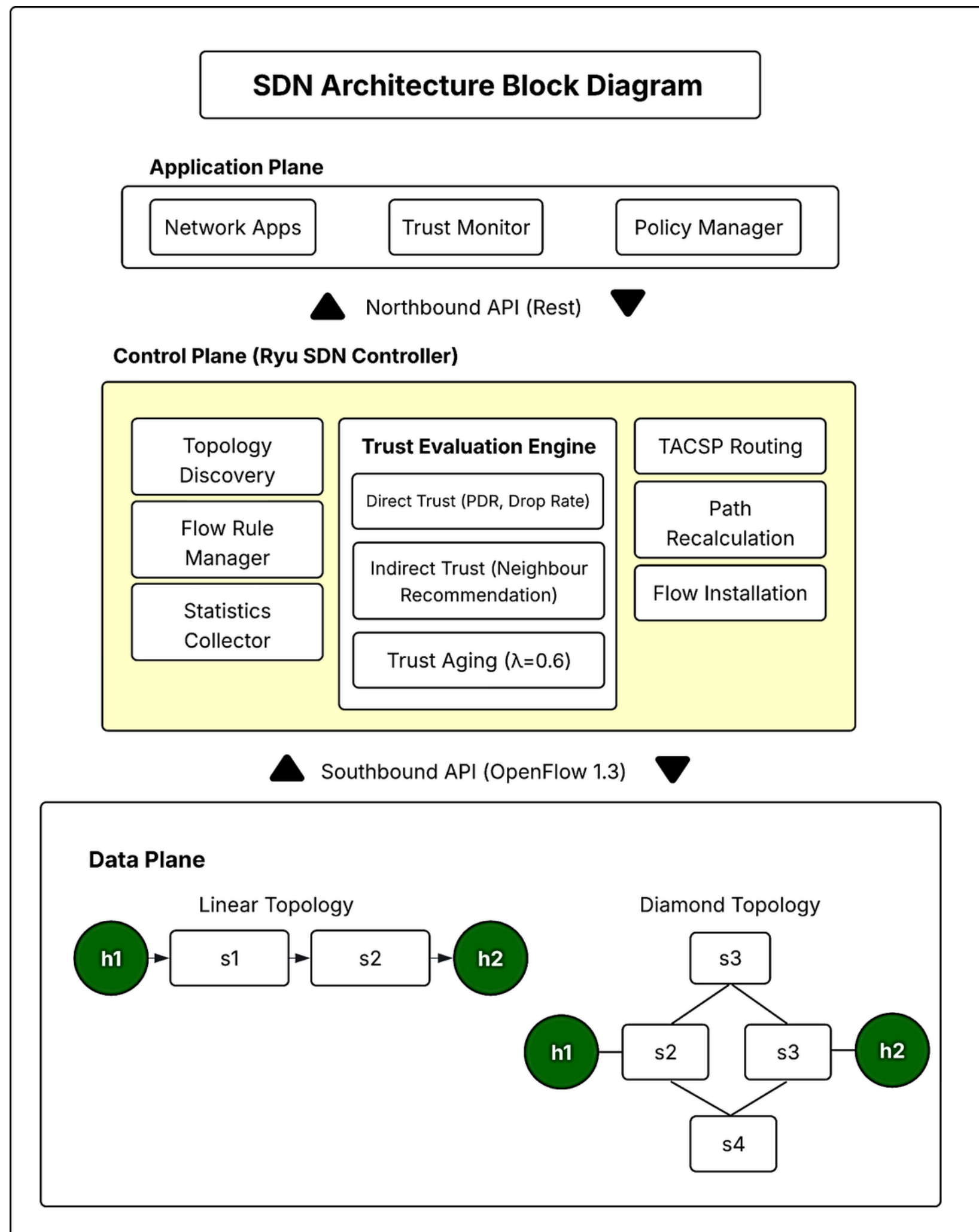
Research Gap & Our Contribution

Gaps in Existing Literature	Our Solution
No routing enforcement	TACSP Routing
Reactive detection only	Early Attack Detection
No trust aging or indirect trust	Hybrid Trust Model
Domain-specific (IoT / 5G / WSN)	Standard SDN (OpenFlow 1.3)

Trust Aware Path Selection Algorithm



SDN Architecture Block Diagram



Trust Model

Direct Trust

$$T_{\text{direct}}(v) = \alpha \cdot \text{PDR}(v) + \beta \cdot D(v) + \gamma \cdot L(v)$$

where $\alpha = 0.5$

Indirect Trust

$$T_{\text{indirect}}(v) = \sum T(u) \cdot T_{\text{reported}}(v) / \sum T(u)$$

Neighbour-weighted reputation

Hybrid Trust

$$T_{\text{hybrid}}(v) = \delta \cdot T_{\text{direct}}(v) + (1 - \delta) \cdot T_{\text{indirect}}(v)$$

$\delta = 0.7$ | Routable only if $T(v) \geq 0.5$

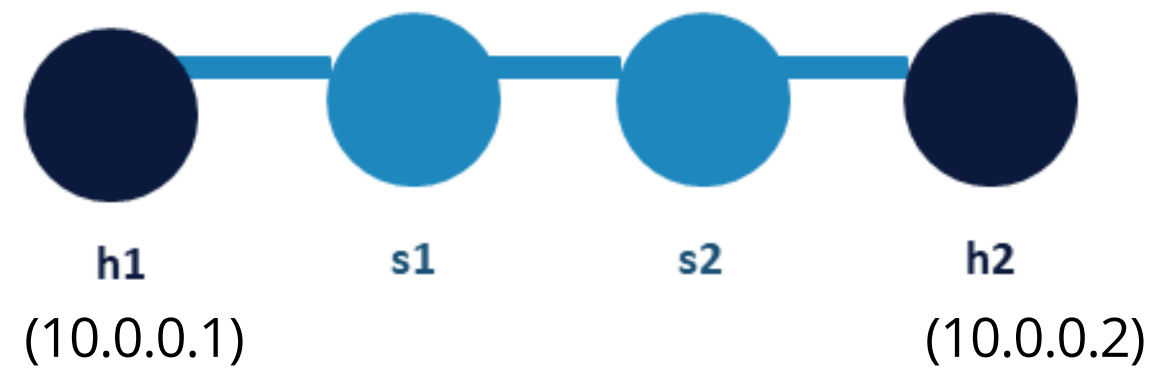
Final Trust

$$T_{\text{final}}(v) = \lambda \cdot T_{\text{hybrid}}(v) + (1 - \lambda) \cdot T_{\text{prev}}(v)$$

$\lambda = 0.6$ | Historical Value contributes 40%

Network Topologies for Evaluation

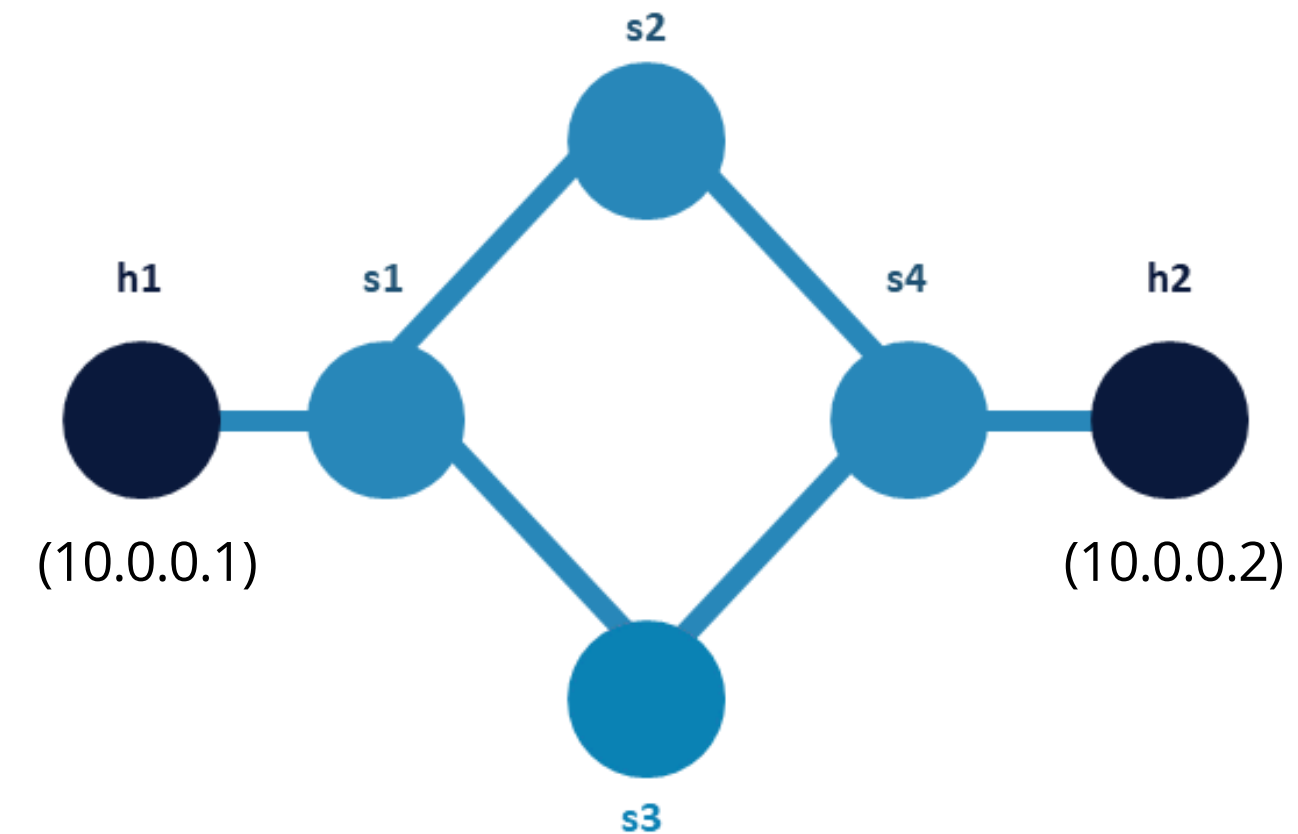
Linear Topology



Basic 2-hop path — attack detection baseline

$h1 \rightarrow s1 \rightarrow s2 \rightarrow h2$

Diamond Topology



Dual-path — reroutes via s2 on s3 attack

Attack Detection Mechanism

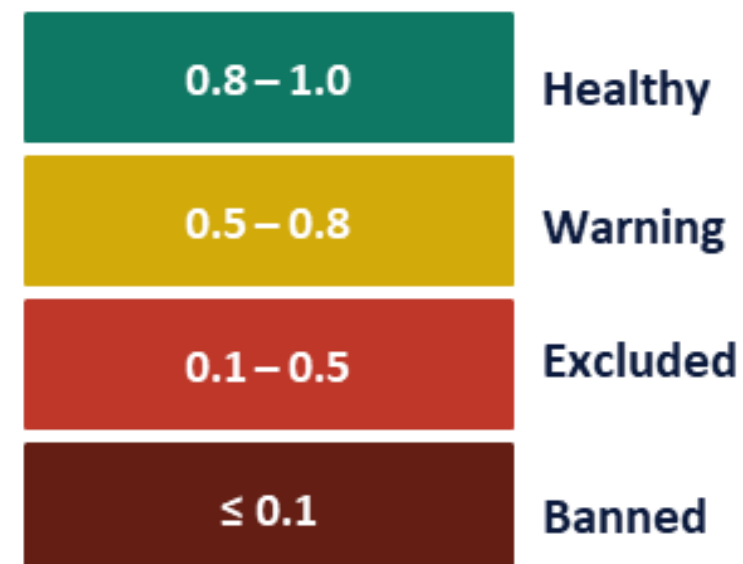
Blackhole Attack

A compromised switch silently drops ALL packets

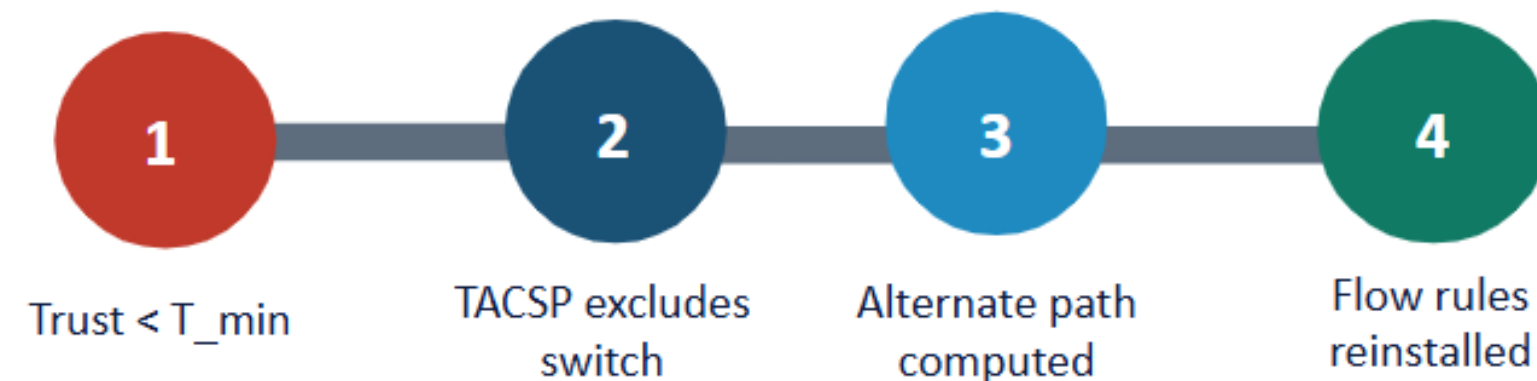
Greyhole Attack

A compromised switch drops only a percentage of packets

Trust score zones



Automated response pipeline



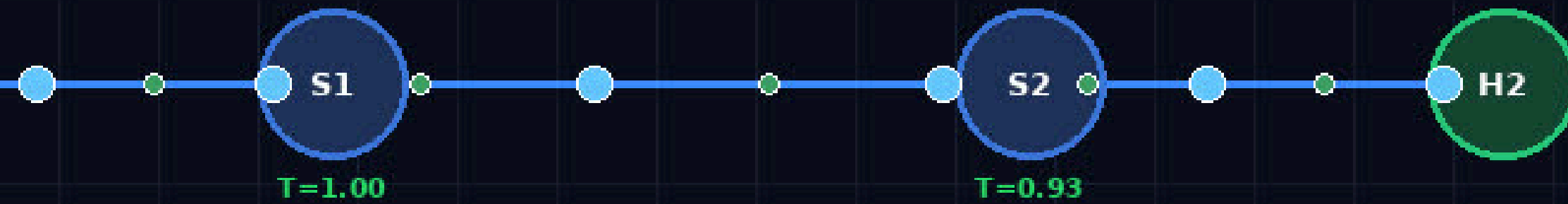
SCENARIO 1 — Normal Operation

Network Stats

Time	10.0s
Packets Sent	60
PDR	90%
Latency	2.1ms
Loss Rate	0%

Trust History

s1	1.00
s2	0.93



✓ Trust stable at 1.0

SCENARIO 2 — Blackhole Attack + Auto-Reroute

Network Status

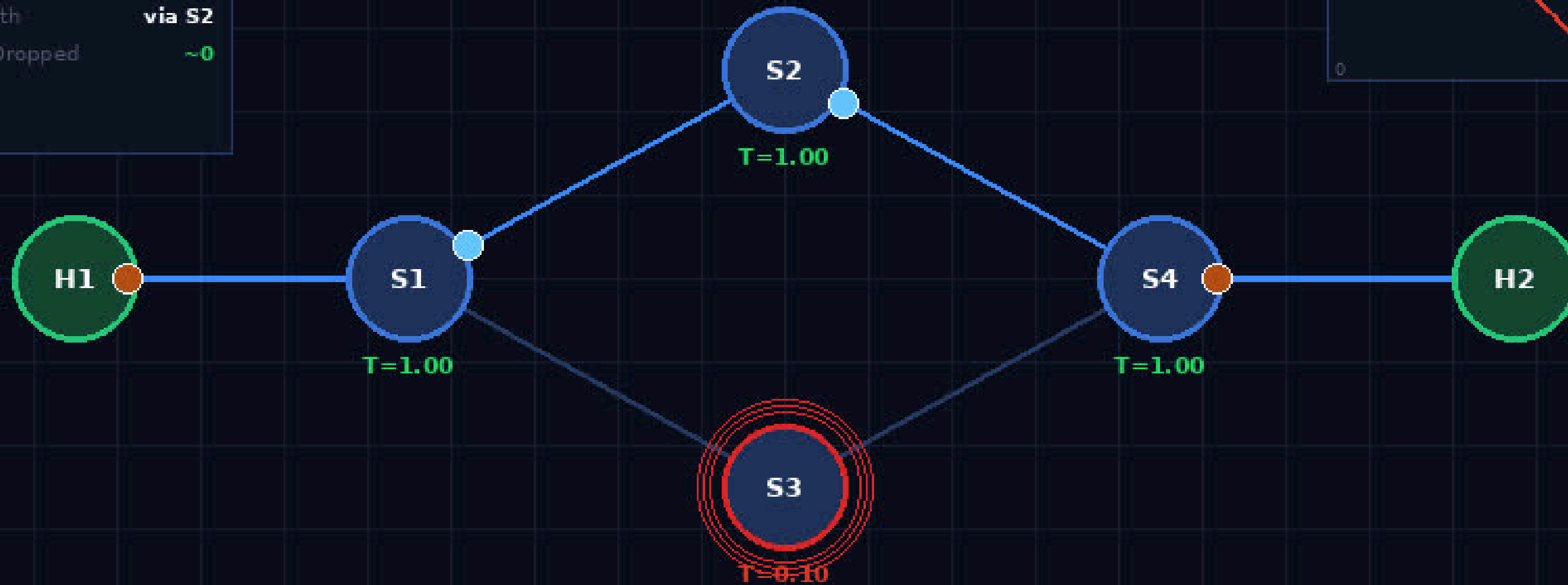
Phase ⚡ **REROUTING via S2**

PDR **93%**

S3 Trust **0.10**

Active Path **via S2**

Packets Dropped **~0**

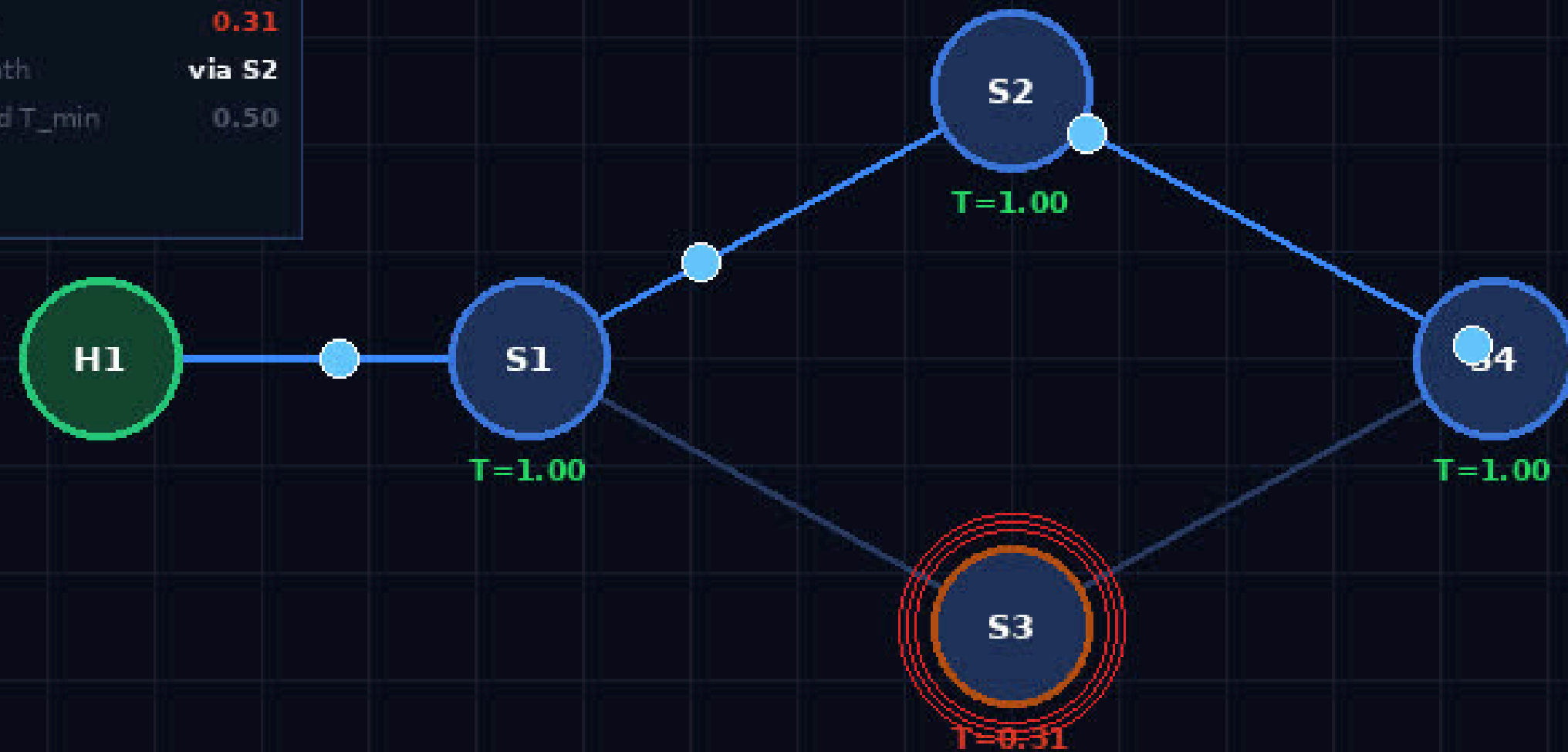


⚡ Rerouting via S2 — S3 isolated (Trust=0.10)

SCENARIO 3 — Greyhole Attack (50% Random Drop)

Network Status

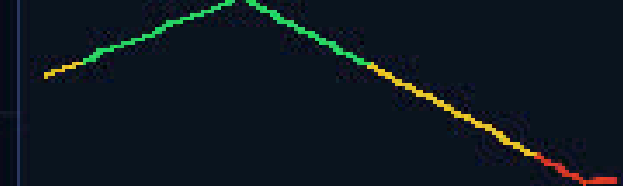
Phase ⚡ **REROUTING via S2**
PDR **78%**
Drop Rate **0%**
S3 Trust **0.31**
Active Path **via S2**
Threshold T_{min} 0.50



PDR Over Time



S3 Trust



● Forwarded

● Dropped

⚡ Trust below $T_{min}=0.5$ — Rerouting via S2

Linear Topology - Blackhole Attack

raj@raj: -

raj@raj: ~ 60x13

[TRUST] s1 | fwd= 151 d= 14 | drop=no | trust=1.000 (was 1.000) | offenses=0 | FORWARDING

[TRUST] s2 | fwd= 149 d= 14 | drop=no | trust=1.000 (was 1.000) | offenses=0 | FORWARDING

[TRUST] s1 | fwd= 167 d= 16 | drop=no | trust=1.000 (was 1.000) | offenses=0 | FORWARDING

[TRUST] s2 | fwd= 165 d= 16 | drop=no | trust=1.000 (was 1.000) | offenses=0 | FORWARDING

[TRUST] s1 | fwd= 181 d= 14 | drop=no | trust=1.000 (was 1.000) | offenses=0 | FORWARDING

[TRUST] s2 | fwd= 179 d= 14 | drop=no | trust=1.000 (was 1.000) | offenses=0 | FORWARDING

raj@raj: ~ 60x13

Scenario 2 - Blackhole Attack

Switches: s1 s2 (recommended: s1)

Switch Trust Scores:

S1

[Progress Bar]

1.000

HEALTHY

offenses=0

S2

[Progress Bar]

1.000

HEALTHY

offenses=0

Which switch to attack?

MiniNAM — LINEAR

64 bytes from 10.0.0.2: icmp_seq=67 ttl=64 time=0.153 ms

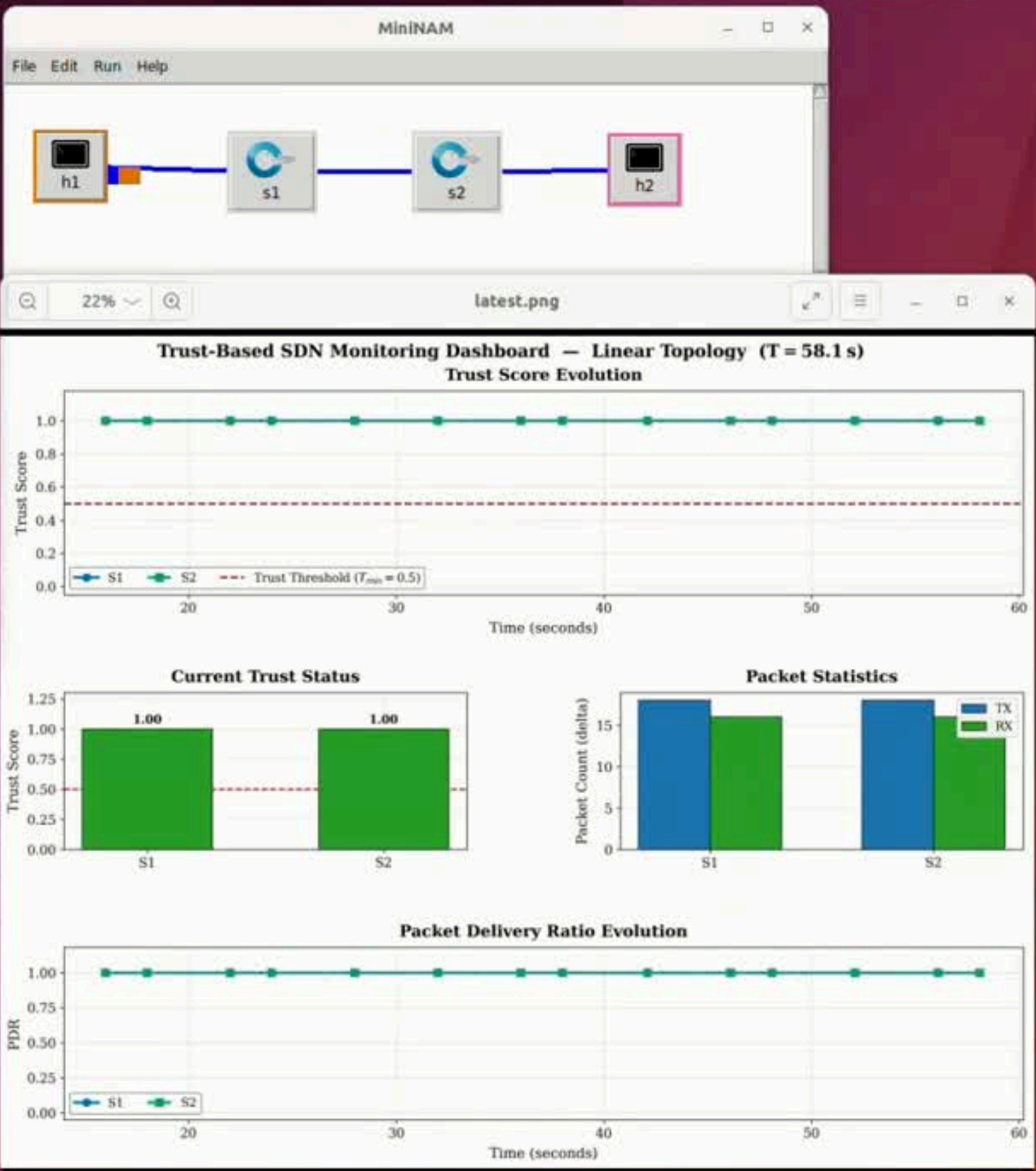
64 bytes from 10.0.0.2: icmp_seq=68 ttl=64 time=0.073 ms

64 bytes from 10.0.0.2: icmp_seq=69 ttl=64 time=0.161 ms

64 bytes from 10.0.0.2: icmp_seq=70 ttl=64 time=0.111 ms

64 bytes from 10.0.0.2: icmp_seq=71 ttl=64 time=0.134 ms

64 bytes from 10.0.0.2: icmp_seq=72 ttl=64 time=0.063 ms



Linear Topology - Greyhole Attack

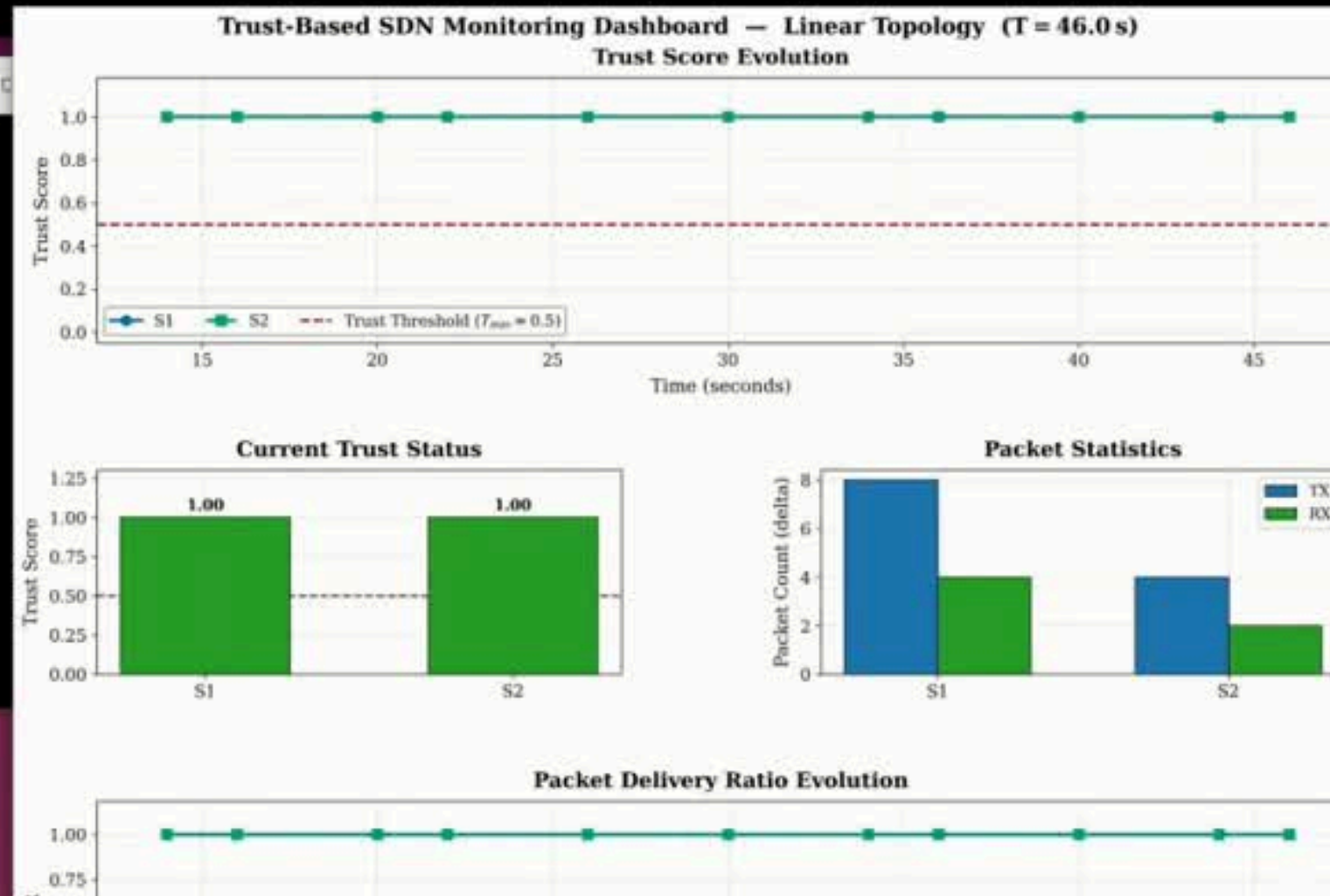
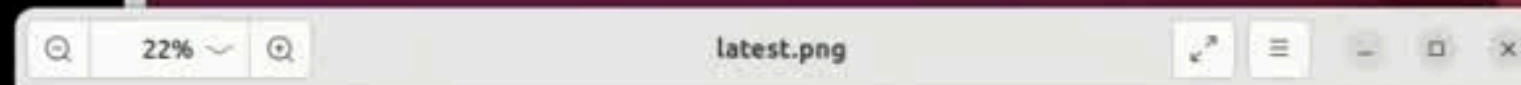
```
raj@raj: ~  
raj@raj: ~ 58x9  
[TRUST] s2 | fwd= 87 d= 16 | drop=no | trust=1.000 (wa  
s 1.000) | offenses=0 | FORWARDING  
[TRUST] s1 | fwd= 89 d= 16 | drop=no | trust=1.000 (wa  
s 1.000) | offenses=0 | FORWARDING  
[TRUST] s1 | fwd= 104 d= 15 | drop=no | trust=1.000 (wa  
s 1.000) | offenses=0 | FORWARDING  
[TRUST] s2 | fwd= 102 d= 15 | drop=no | trust=1.000 (wa  
s 1.000) | offenses=0 | FORWARDING  
[
```

SCENARIO MENU [LINEAR]

- 1) Normal Operation - observe healthy traffic
- 2) Blackhole Attack - drop ALL packets
- 3) Greyhole Attack - drop PARTIAL packets
- 4) Repeat Attacker - 4 cycles → permanent ban
- 5) Show Trust Scores
- 6) Open Graph
- 7) Stop Attack
- 8) Exit

Select [0-7]:

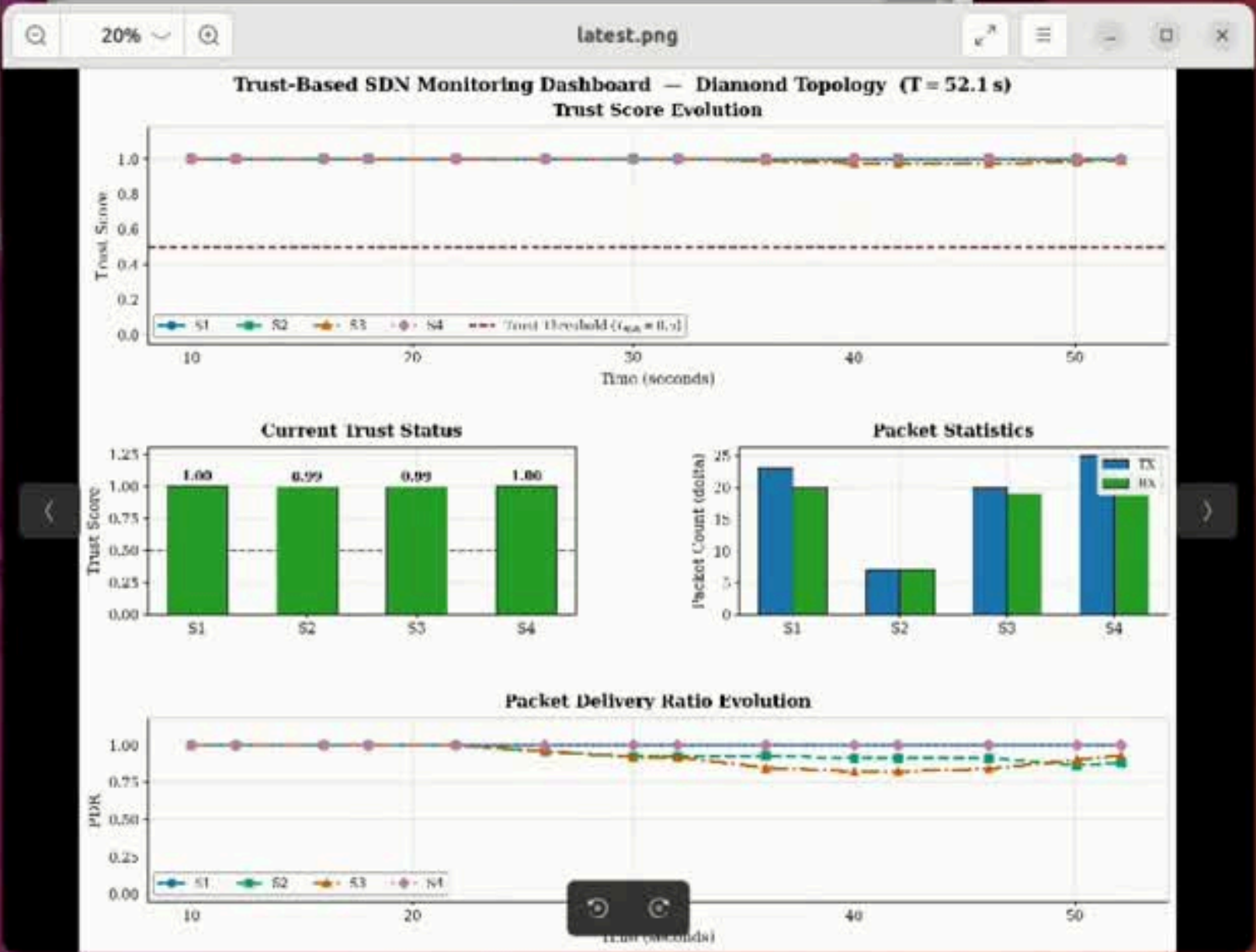
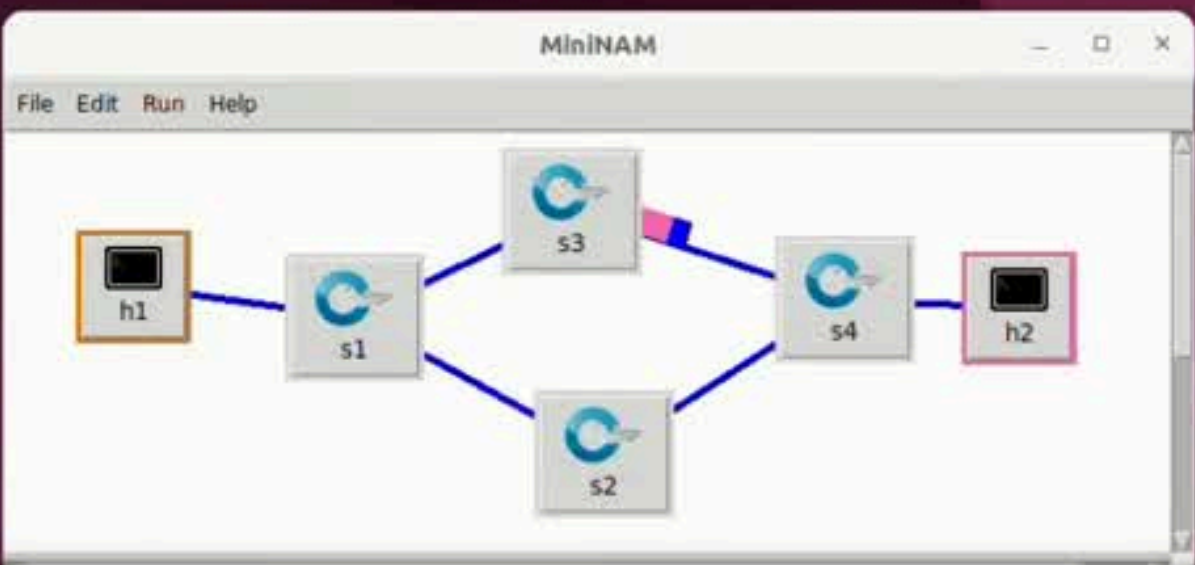
```
MinINAM — LINEAR  
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.080 ms  
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.095 ms  
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.217 ms  
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.170 ms  
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.054 ms  
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.074 ms  
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.124 ms  
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.062 ms  
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.159 ms  
64 bytes from 10.0.0.2: icmp_seq=27 ttl=64 time=0.115 ms  
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.055 ms  
64 bytes from 10.0.0.2: icmp_seq=29 ttl=64 time=0.154 ms  
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=0.134 ms  
64 bytes from 10.0.0.2: icmp_seq=31 ttl=64 time=0.153 ms  
64 bytes from 10.0.0.2: icmp_seq=32 ttl=64 time=0.139 ms  
64 bytes from 10.0.0.2: icmp_seq=33 ttl=64 time=0.069 ms  
64 bytes from 10.0.0.2: icmp_seq=34 ttl=64 time=0.082 ms  
64 bytes from 10.0.0.2: icmp_seq=35 ttl=64 time=0.064 ms  
64 bytes from 10.0.0.2: icmp_seq=36 ttl=64 time=0.129 ms  
64 bytes from 10.0.0.2: icmp_seq=37 ttl=64 time=0.109 ms  
64 bytes from 10.0.0.2: icmp_seq=38 ttl=64 time=0.090 ms  
[
```



Diamond Topology - Blackhole Attack

```
raj@raj: ~  
raj@raj: ~$ 55x9  
[TRUST] s1 | fwd= 176 d= 18 | drop=no | trust=1.000  
(was 1.000) | offenses=0 | FORWARDING  
[TRUST] s3 | fwd= 176 d= 18 | drop=no | trust=1.000  
(was 1.000) | offenses=0 | FORWARDING  
[TRUST] s2 | fwd= 88 d= 4 | drop=no | trust=0.991  
(was 0.992) | offenses=0 | FORWARDING  
[TRUST] s4 | fwd= 176 d= 19 | drop=no | trust=1.000  
(was 1.000) | offenses=0 | FORWARDING  
[  
  
raj@raj: ~$ 55x12  
SCENARIO MENU [DIAMOND]  
1) Normal Operation - observe healthy traffic  
2) Blackhole Attack - drop ALL packets  
3) Greyhole Attack - drop PARTIAL packets  
4) Repeat Attacker - 4 cycles → permanent ban  
5) Show Trust Scores  
6) Open Graph  
7) Stop Attack  
8) Exit  
Select [0-7]:
```

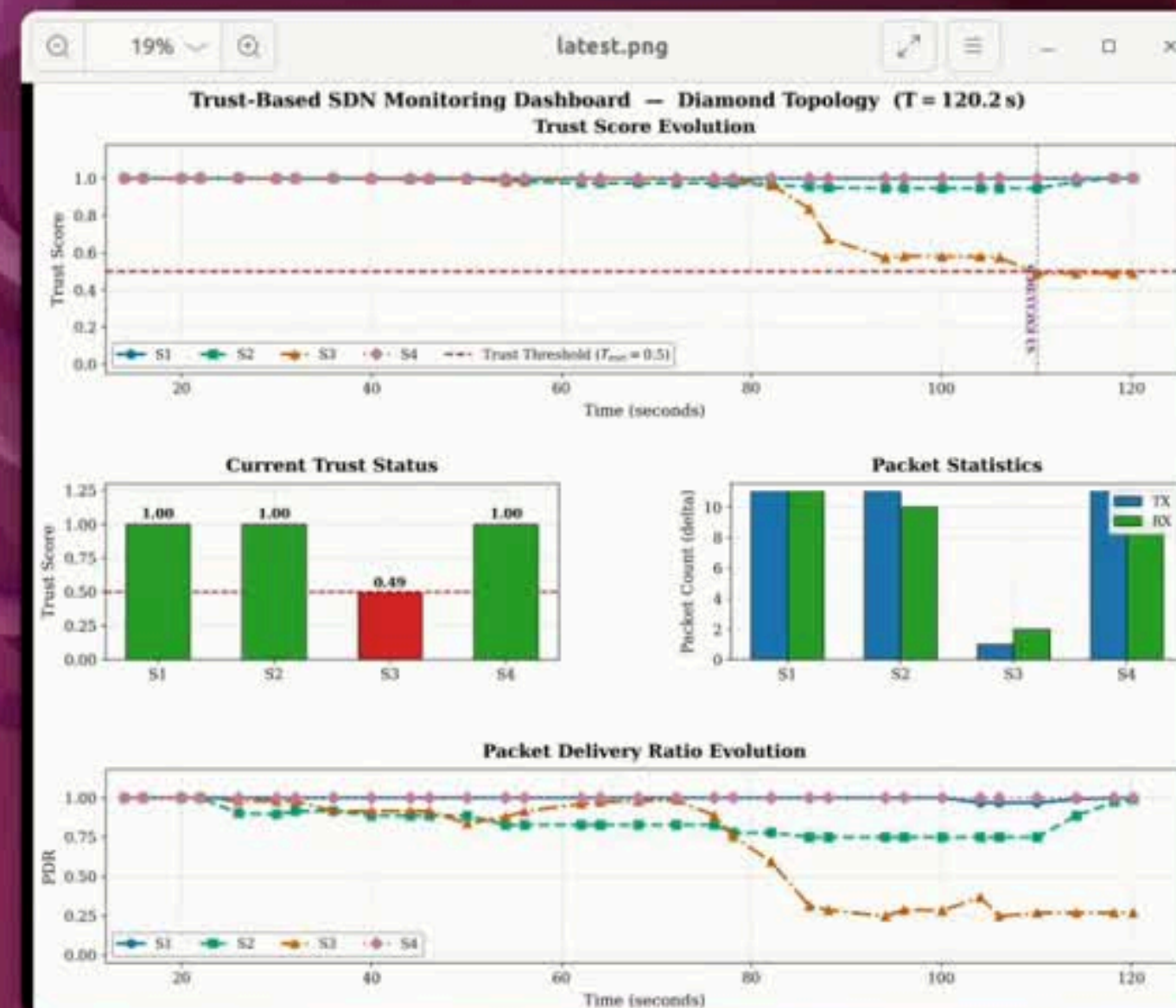
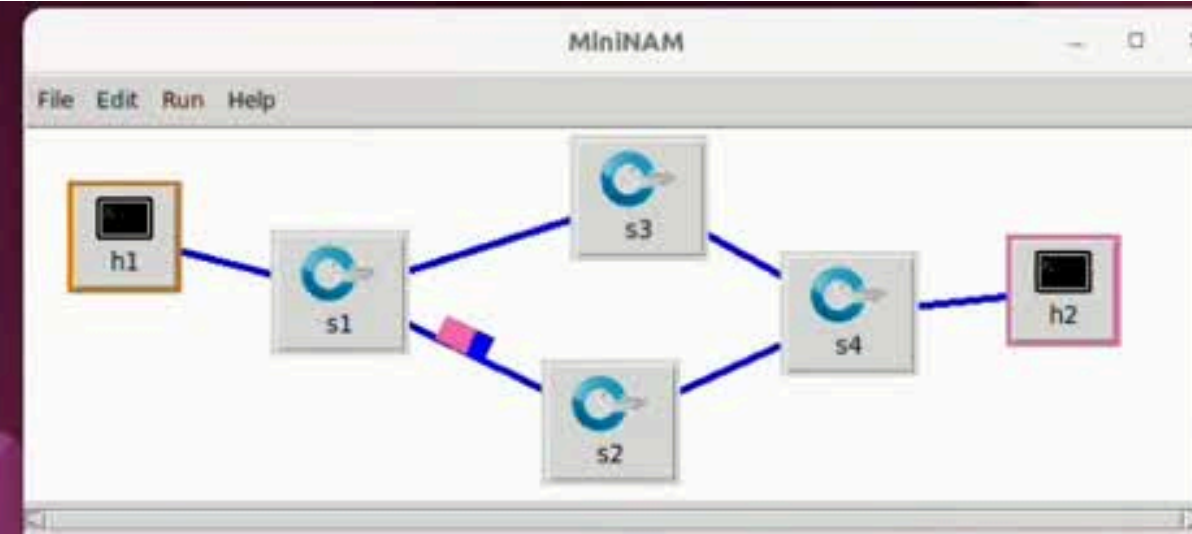
```
MiniNAM — DIAMOND  
80 ms  
64 bytes from 10.0.0.2: icmp_seq=40 ttl=64 time=0.1  
51 ms  
64 bytes from 10.0.0.2: icmp_seq=41 ttl=64 time=0.1  
59 ms  
64 bytes from 10.0.0.2: icmp_seq=42 ttl=64 time=0.1  
71 ms  
64 bytes from 10.0.0.2: icmp_seq=43 ttl=64 time=0.1  
41 ms  
64 bytes from 10.0.0.2: icmp_seq=44 ttl=64 time=0.0  
82 ms  
64 bytes from 10.0.0.2: icmp_seq=45 ttl=64 time=0.0  
83 ms  
64 bytes from 10.0.0.2: icmp_seq=46 ttl=64 time=0.1  
30 ms  
64 bytes from 10.0.0.2: icmp_seq=47 ttl=64 time=0.1  
54 ms  
64 bytes from 10.0.0.2: icmp_seq=48 ttl=64 time=0.2  
17 ms  
64 bytes from 10.0.0.2: icmp_seq=49 ttl=64 time=0.1  
76 ms  
64 bytes from 10.0.0.2: icmp_seq=50 ttl=64 time=0.1  
28 ms  
[
```



Diamond Topology - Greyhole Attack

```
raj@raj: ~  
raj@raj: ~ 50x11  
[TRUST] s1 | fwd= 280 d= 7 | drop=no | trust=1  
.000 (was 1.000) | offenses=0 | FORWARDING  
[TRUST] s4 | fwd= 276 d= 7 | drop=no | trust=1  
.000 (was 1.000) | offenses=0 | FORWARDING  
[TRUST] s2 | fwd= 318 d= 8 | drop=no | trust=1  
.000 (was 1.000) | offenses=0 | FORWARDING  
[QUARANTINE] s3 PERMANENTLY BANNED - recovery bloc  
ked forever  
[TRUST] s3 | fwd= 230 d= 2 | drop=no | trust=0  
.489 (was 0.489) | offenses=16 | FORWARDING  
raj@raj: ~ 50x11  
S3 [ ] 1.000 HEALTHY offe  
nses=0  
S4 [ ] 1.000 HEALTHY offe  
nses=0  
Which switch to attack? s3  
Drop % [30/50/70]: 70  
Injecting GREYHOLE on s3 (70% drop)...  
Attack ACTIVE - dots SPARSE in MiniNAM  
Press ENTER when ready to STOP attack...
```

```
MiniNAM — DIAMOND  
=0.137 ms  
64 bytes from 10.0.0.2: icmp_seq=239 ttl=64 time  
=0.060 ms  
64 bytes from 10.0.0.2: icmp_seq=240 ttl=64 time  
=0.065 ms  
64 bytes from 10.0.0.2: icmp_seq=241 ttl=64 time  
=0.081 ms  
64 bytes from 10.0.0.2: icmp_seq=242 ttl=64 time  
=0.072 ms  
64 bytes from 10.0.0.2: icmp_seq=243 ttl=64 time  
=0.071 ms  
64 bytes from 10.0.0.2: icmp_seq=244 ttl=64 time  
=0.167 ms  
64 bytes from 10.0.0.2: icmp_seq=245 ttl=64 time  
=0.137 ms  
64 bytes from 10.0.0.2: icmp_seq=246 ttl=64 time  
=0.155 ms
```



Communication and Protocols

Layer Name	Protocol	Purpose
Application	ICMP (Ping)	Tests connectivity between h1 and h2
Transport	TCP	Controller to Switch communication
Network	IPv4	Logical addressing
Data Link	Ethernet	MAC addressing within the network
Control Plane	OpenFlow 1.3	Manage and Control switch forwarding rule
Discovery	LLDP	Switch-to-switch link discovery
Address Resolution	ARP	Resolves IP to MAC address

Real - World Applications

Critical Infrastructure

- Power grids
- Water systems
- Smart cities

Data Centre Networks

- Cloud
- Hyperscale
- Multi-tenant

5G / Next-Gen Networks

- SDN-enabled RAN
- Core slicing

Enterprise & Campus LANs

- Hospitals
- Universities
- Govt. networks

IoT & Industrial Networks

- IIoT
- SCADA
- Smart manufacturing

Military & Defence MANETs

- Tactical SDN
- Secure communications

Conclusion

Problem Addressed

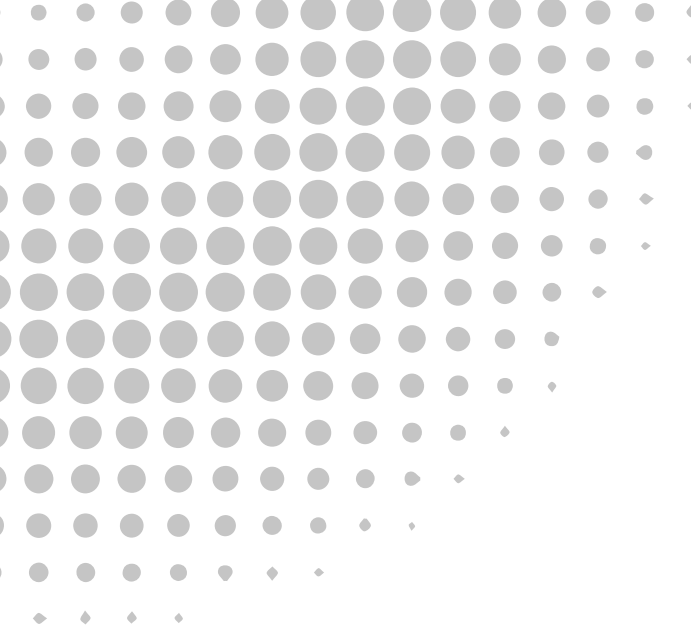
- Traditional OpenFlow networks blindly trust switches
- Blackhole and greyhole attacks go undetected
- Trust is evaluated but not used in routing decisions

Our Approach

- Introduced a hybrid trust model (direct + indirect + aging)
- Integrated trust into routing using TACSP with Dijkstra
- Enabled automatic path updates when trust drops

Future Work

- Scale to larger networks with multiple controllers
- Use machine learning for dynamic trust tuning
- Explore blockchain-based security and real-world testing



Thank You !

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.

